SYED IBAD HASNAIN

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Registered Biomedical Engineer (Pakistan Engineering Council License No. Biomedical/002773)

PROFILE SUMMARY

Research-oriented Biomedical Engineer with a strong foundation in Deep Learning, Machine Learning, and AI-driven biomedical applications. Skilled in developing data-driven models and applying computational methods to address complex healthcare challenges. Seeking a PhD/RA opportunity where I can contribute to impactful research, collaborate with interdisciplinary teams, and support the development of innovative solutions that advance healthcare and intelligent technologies.

KEY ACHIEVEMENTS

- Secured Merit-based Scholarships throughout undergraduate and postgraduate studies.
- Registered Scientist — National Scientists Directory (NSD), PASTIC, Ministry of Science & Technology, Govt. of Pakistan (NSD-MultiDisc-0012428)
- Secured National Funding by IGNITE (Ministry of IT) on our final year design project "*Portable Health tracking system for COVID19 Era*". Project Code: *NGIRI-2021-9948*
- Serving as Reviewer for International Journals (Biosensors & Bioelectronics and Current Psychology).
- Elected and Served as Vice-chairperson IEEE EMBS SSUET Student Chapter, Karachi, Pakistan. Membership No. 98015228, (2021-2022)
- Finalist (Ranked 2nd) – MSCA COFUND PhD Programme, University of Bologna, Italy (2025)
- Qualified for PhD in Bioengineering – Politecnico di Milano (Ranked 41st globally) (2025)

EDUCATION

Postgraduate Diploma in Data Science with Artificial Intelligence
NED University of Engineering & Technology (*Jul 2024 – Dec 2025)*

Masters of Science in Biomedical Engineering
Sir Syed University of Engineering & Technology (*Sep 2022 – Oct 2025)* ***cGPA: 3.58/4.0***

Bachelors of Science in Biomedical Engineering
Sir Syed University of Engineering & Technology (*Jan 2018 – Mar 2022)* ***cGPA: 3.27/4.0***

RESEARCH INTERESTS

Digital Image and signal Processing, AI for Healthcare & Clinical Decision Support (Multimodality & Explainable in AI), Deep Learning, NLP and Agentic AI.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, R, SQL, C++.
- **Machine Learning:** Regression, Classification, Clustering; Model Evaluation, Cross-validation
- **Deep Learning:** CNNs, RNNs, LSTM; Time-Series & Sequence Modeling
- **Libraries & Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy
- **Computer Vision:** OpenCV, Roboflow
- **AI Tools & Deployment:** Hugging Face, Chainlit
- **Simulation & Circuit Tools:** COMSOL Multiphysics, Proteus, Multisim
- **Embedded Systems / IoT:** Raspberry Pi, Arduino, ESP32
- **Automation & Workflow:** n8n
- **Technical Writing:** LaTeX, Mendeley, EndNote

WORK EXPERIENCE

Lecturer, *Hamdard University- (Faculty of Engineering, Sciences and Technology.)* ***(Aug 2025 – Present)***

- Deliver lectures and lab sessions in Biomedical Engineering courses including (BME-422 Medical Image Processing, EE-221 Digital Logic Design, BME-326 Telemedicine Systems and CSOP-3202 Objected Oriented Programming).

- Supervise undergraduate student projects in AI and biomedical applications.
- Contribute to departmental research initiatives in deep learning for healthcare.

Cambridge IGCSE/GCE Educator, *The Next School-Alpha Education Network* (***Aug 2024 – July 2025)***

- Taught Computer Science, Physics, and Additional Mathematics to IGCSE/GCE students.
- Designed and delivered STEM-focused curriculum integrating computational thinking.

Mathematics & Robotics Head,
Army Public School & College System (***Sep 2022 – Jul 2024***)

- Led robotics club activities and trained students in robotics and programming.
- Conducted teacher training sessions and contributed to curriculum development.

Biomedical Engineer, *Radiant Medical Pvt Ltd (**Jan 2022- Aug 2022)***

- Performed installation, calibration, and maintenance of medical equipment across critical care and radiology departments
- Gain hands-on training on numerous medical equipment: Ultrasound, X-ray, Ventilator, Defibrillator, ECG and Anesthesia machine.

Intern, *Sindh Institute of Urology and Transplantation* (***Jan 2021 – Feb 2021***)

- Attended workshops that highlighted different techniques to resolve malfunctioning issues in medical devices.
- Worked under the supervision of a biomedical engineer to grasp the techniques of installation, calibration, and maintenance of equipment.

PROJECTS

Department of Computer Science & Information Technology, NEDUET, Karachi, Pakistan.
"A Hybrid Deep Learning Approach for Multi-Class Classification of Renal Lesions in CT Imaging."

- Developed a clinically deployable AI pipeline for multi-class renal lesion classification in CT (normal, cysts, stones, tumors), using a hybrid approach combining lightweight CNNs with advanced modules (KELM, DBN, RNNs, Vision Transformers). Built a reproducible workflow with dataset cleaning, stratified splitting, imbalance handling (SMOTE), and clinical visualization. Benchmarking across 13+ metrics showed KELM + SMOTE achieved 99.71% accuracy, outperforming complex deep architectures and supporting efficient, interpretable diagnostic decision support.

Department of Computer Science & Information Technology, NEDUET, Karachi, Pakistan. "Biomedical Device Troubleshooting Assistant"

- Designed a safety-constrained agentic LLM system for biomedical device troubleshooting that emphasizes clinical engineering workflow alignment, interpretability, and safe deployment. Integrated tool-augmented reasoning with strict policy enforcement (no clinical advice, invasive repair, alarm bypass, firmware tampering, or PHI handling), producing structured troubleshooting, QC guidance, and documentation templates. The assistant demonstrates how guardrailed agentic AI can support reliable decision support in healthcare-adjacent settings.

Department of Biomedical Engineering, SSUET, Karachi, Pakistan.
"Evaluation and Implementation of Machine Learning Algorithms for Prediction of Kidney and Heart Diseases in Diabetic Patients."

- Built a machine learning pipeline to predict CKD and CVD risk in diabetic patients using Pakistan's clinical dataset (n=703). Statistical feature validation (SPSS) and ML benchmarking showed Random Forest achieved 92% accuracy, supporting early detection of silent complications in low-resource healthcare settings.
- Hasnain, S. I. (2025). Evaluation and Implementation of Machine Learning Algorithms to Predict Early Detection of Kidney and Heart Disease in Diabetic Patients. *arXiv preprint arXiv:2510.14997*.

Department of Biomedical Engineering, SSUET, Karachi, Pakistan.
"Portable health tracking system for COVID19 Era"

- Developed a real-time patient monitoring system using SpO2, temperature, heart rate, and respiratory sensors connected to a microcontroller. Data was displayed locally and transmitted to a mobile app for remote tracking by healthcare providers. Enabled centralized, real-time health monitoring to improve patient care during the pandemic.

PUBLICATIONS

- **Hasnain, S. I**., & Saleem, D. (2025). Impact of aging on glycemic control: A gender-differentiated analysis of random blood sugar levels. *Rawal Medical Journal*, *50*(3), 569-569. ***[Scopus, WoS , ESCI (IF 0.7) Q4]***

- **Hasnain, S. I.**, Israr, H., Faris, M., Kamal, R., & Tirmizi, H. S. Y. (2025). Evaluation of Machine Learning–Based Methods to Detect Bipolar Disorder in Individuals With Mental Health Conditions. *VFAST Transactions on Software Engineering*, *13*(3), 129-139. ***[Scopus Indexed]***
- Faris, M., Khalid, S., **Hasnain, S. I.**, Jamil, M., Yasmeen, H., & Arif, M. (2025). Scalable Hybrid Deep Learning-Based Architecture for Glaucomatous and Healthy Eye Classification in Retinal Fundus Images. *VFAST Transactions on Software Engineering*, *13*(4), 01-12. ***[Scopus Indexed]***
- **Hasnain, S. I.**, Faris, M., Tirmizi, H. S. Y., Kamal, R., & Javid, T. (2025). Simulation-Based Machine Learning Approach to Classify Accelerated Biological Aging Using Telomere and Epigenetic Markers. *Sir Syed University Research Journal of Engineering & Technology*, *15*(2), 23-31.
- Faris, M., Khalid, F., Shahid, M., Haque, Z. U., **Hasnain, S. I.,** & Ali, M. F. (2026). An Ensemble Deep Learning Framework for Automated Multi Class Skin Lesion Classification Using ConvNeXt-Tiny, EfficientNetV2-S, and MobileNetV3. *VFAST Transactions on Software Engineering*, *14*(1), 60-72.

CONFERENCES

- **Hasnain, S. I.,** Saleem, D., Israr, H. & Syed, S. A. "A Machine Learning Approach for Risk Stratification of Cardiovascular Disease in Type 2 Diabetic Cohort." *2026 Global Conference on Wireless and Optical Technologies (GCWOT)*. IEEE, Malaga, Spain, 2026. **(Accepted, to appear in IEEE Xplore)**
- Israr, H., Saleem, M. F., Syed, S. A., **Hasnain, S. I.,** & Saleem, D. "BENCHMARKING SUPERVISED AND UNSUPERVISED MACHINE LEARNING MODELS FOR EARLY DETECTION OF BREAST CANCER " *2026 Global Conference on Wireless and Optical Technologies (GCWOT)*. IEEE, Malaga, Spain, 2026. **(Accepted, to appear in IEEE Xplore)**
- **Hasnain, Syed Ibad,** Faris, Muhammad, Crescenzo Pepe, Mir Farooq Ali, and Silvia Maria Zanoli. "AI-Driven Ensemble Approaches for Early Prediction of Cardiovascular Disease Risk " In *2026 27th International Carpathian Control Conference (ICCC)*, IEEE, Miskolc-Szilvásvárad, Hungary, 2026. **(Under review)**
- **Hasnain, Syed Ibad,** Rabail Khowaja**,** Faris, Muhammad, Crescenzo Pepe, Mir Farooq Ali, and Silvia Maria Zanoli. "Hybrid MobileNetV2–Vision Transformer Framework for Automated Multi-Class Renal Lesion Classification in CT Imaging", *2026 34th Mediterranean Conference on Control and Automation (MED)*, IEEE, Ancona, Italy, 2026. **(Under review)**

HONORS & RECOGNITION

- Received Appreciation Certificate for Outstanding Teaching Performance – Army Public School & College System (2023)
- Received Appreciation Certificate for Active Teacher Participation in Co-Curricular Activities – Army Public School & College System (2024)
- Received Appreciation Certificate for conducting Python coding class in Summer Camp – The Next School (2025)
- Received Appreciation Certificate for conducting a workshop on "Design Thinking for Educators" – The Next School (2025)
- Received Recognition for STEM Event Coordination – Alpha Education Network (2024)
- Received Recognition Certificate as THE NEXT SCHOOL Coordinator, Pakistan – International kangaroo Mathematics Contest, (2024)
- Received Appreciation Letter for Outstanding Teaching Performance – The Next School (2025)

CERTIFICATION & TRAINING

- Deep Learning Specialization – Coursera (Andrew Ng)
- Python for Machine Learning – IBM (Coursera)
- Cybersecurity Fundamentals – Google
- Data Analysis Using Python – 10Pearls University
- Object-Oriented Programming (Advanced) – 10Pearls University
- IoT System Development & Applications – NAVTAC (Government of Pakistan)
- Received training on "Design Thinking" at Alpha Education Network, Karachi, Pakistan.
- Received hands on training on "Power Lab Equipment" at Faculty of Health and Medical Science, Hamdard University, Pakistan.

REFERENCES

AVAILABLE UPON REQUEST

PORFOLIO/EVIDENCE: Click to view all documents